

Bitcoin vs The Moon

Claude Autopilot Build Specification v2 - Existing Python Codebase + LiteSpeed/MySQL

MASTER INSTRUCTION: You are working inside the existing local repository for github.com/kandekore/moonvsbtc. This is an existing Python project with working lunar/Bitcoin research logic. Extend and reorganise it; do not throw away provenance-sensitive research code. The production server is LiteSpeed-based and MySQL is available. Optimise for a fast, coherent v1 that can be deployed on that environment. There is currently no meaningful audience, so make sensible decisions autonomously and iterate rather than blocking on perfection.

1. Mission and Autopilot Rules

Act as lead engineer, product architect, data/research engineer and implementation agent. Inspect the entire repository before changing architecture. Check git status, dependencies, current deployment files, .streamlit configuration, app.py, moon_engine.py, btc_moon_analysis.py, btcvsmoon.py, requirements.txt, Dockerfile and all supporting files. Create a safe git checkpoint/branch, write a concise plan, then IMPLEMENT it immediately.

Do not repeatedly ask the user about routine engineering choices. Only stop for a genuinely blocking credential, destructive production action, paid service commitment, or ambiguity that cannot safely be resolved. Prefer working vertical slices over elaborate architecture. Run the application, tests, lint/type checks where available, and fix critical failures as you go.

Never commit secrets. Use environment variables and provide .env.example. Preserve git history and make logical commits/checkpoints if practical.

2. What the Product Is

Bitcoin vs The Moon is a public longitudinal experiment investigating whether lunar cycles and Bitcoin's astrological natal chart show useful relationships with Bitcoin market behaviour when considered alongside conventional technical analysis, price structure, volume, market regime and real-world catalysts.

It is NOT a generic crypto-news portal and NOT a claim that the Moon reliably predicts Bitcoin. The central proposition is: "Can lunar cycles tell us anything useful about Bitcoin? We are testing the hypothesis publicly - recording observations before events and then recording what actually happened, including failures."

The human owner is Editor and Chief Scientist. The AI is Researcher, Tester, Journalist and Astrologer. AI gathers evidence, performs calculations, monitors active experiments, drafts analysis and maintains records. The human decides what is published.

3. Private Laboratory vs Public Website

The private Companion is the laboratory. Private chat, screenshots, trade positions, leverage, exchange balances, P&L, stop losses, take profits and personal trading decisions are PRIVATE by default and must never be exposed through public pages or APIs.

Nothing from private chat publishes automatically. The owner explicitly promotes material using actions such as Create Observation, Create Hypothesis, Create Prediction, Draft Article, Record Result, Attach to Experiment or Ignore.

Public material can discuss market hypotheses and lessons, but must not include private trading details unless the owner deliberately inserts them into the public draft.

4. Existing Python Codebase - Preserve Provenance

Treat existing lunar/research modules as provenance-sensitive. In particular, audit moon_engine.py, btc_moon_analysis.py and btcvsmoon.py before refactoring. Preserve the ability to reproduce the legacy website methodology and the historical +4.4-day result.

Do not rewrite working research calculations merely to make the code look cleaner. If refactoring, first add regression tests around the current outputs, then move code behind clear services/modules.

Create PROJECT_CONTEXT.md containing the product vision, research methodology, privacy boundary, editorial rules and deployment notes so future Claude sessions retain context.

5. Recommended Python Architecture

Stay Python-first. Do not migrate to Next.js/TypeScript. The existing Streamlit app may be retained for the private research/admin interface if it accelerates delivery.

For the PUBLIC editorial/SEO site, strongly consider Flask + Jinja templates (or an equally lightweight WSGI Python framework already compatible with the repository) because the production server is LiteSpeed-based, MySQL is available, and a conventional server-rendered site is better suited to articles, canonical URLs, metadata, schema markup and search indexing than a Streamlit-only public site.

A pragmatic v1 is allowed to be hybrid: existing/reworked Streamlit for private Companion/research tools + Flask/Jinja for public pages, sharing the same Python research services and MySQL database. If the current hosting/deployment makes one-process Flask materially simpler, it is acceptable to build the admin/Companion inside Flask as authenticated pages instead. Choose the fastest maintainable route after auditing deployment.

Do not run two frameworks merely for architectural purity. If Streamlit can safely satisfy both private and public requirements on this host, document why and proceed. If it creates SEO/auth/deployment friction, use Flask for public delivery and preserve Streamlit only where it adds value.

6. LiteSpeed + MySQL Deployment Constraints

Production target: LiteSpeed-based hosting with MySQL available. Audit the repository's existing deployment approach before choosing runtime details. Prefer WSGI/Passenger-compatible deployment for Flask if supported by the host. If LiteSpeed is fronting another Python process, document the reverse-proxy/process-manager configuration rather than assuming it.

Use MySQL as the persistent production datastore. Use SQLAlchemy (plus Alembic/Flask-Migrate or equivalent) so schema migrations are versioned. SQLite may be used only for local development/testing if useful; production must be MySQL-ready.

Use a MySQL driver appropriate to the host (for example PyMySQL/mysqlclient based on available build support). Keep DATABASE_URL/credentials in environment variables. Use utf8mb4.

Scheduled research jobs must not depend on a browser session. Implement CLI/job entry points that can be run by server cron. The twice-daily briefings should therefore be executable by commands such as `python -m jobs.morning_brief` and `python -m jobs.evening_brief`, with idempotency/job-run records to prevent duplicates.

Provide DEPLOYMENT.md with exact LiteSpeed/MySQL setup assumptions, environment variables, migration commands, cron commands, static-file handling, restart procedure and troubleshooting notes.

7. Suggested Module Layout

Refactor only as needed, but aim toward clear boundaries such as: `research/` (legacy methodology, scoring, statistical tests), `lunar/` (phase calculations and frozen windows), `astrology/` (BTC natal chart/transits), `market_data/` (price/volume adapters), `news/` (RSS/source ingestion), `ai/` (OpenAI client/prompts/retrieval), `editorial/` (observations/predictions/results/articles), `jobs/` (scheduled research), `models/` (SQLAlchemy), `public/` or `templates/` (public site), `admin/` (private tools), `tests/`.

Keep external providers behind adapters so `market/news/OpenAI` sources can be changed without rewriting the application.

8. Research Constitution - Do Not Rewrite History

The attached/referenced August 2026 research paper is exploratory / hypothesis-generating. Its conclusion is continued testing, not a standalone lunar trading signal.

Legacy website benchmark: BTC-USD Yahoo Finance daily Close; lunar phases generated with PyEphem and reduced to calendar dates; scipy.signal.find_peaks for significant highs/lows; minimum spacing 30 observations; prominence threshold 15% of median Close; each Full Moon matched to nearest significant swing high within +/-14 calendar days; New Moon analogously matched to nearest swing low; signed lag = pivot date minus Moon date.

Recent two-year Full-Moon benchmark reproduced mean +4.4 days, median +4 days, SD about 4.2 days, n=5. Across 2017-2026, roughly 70% of matched major highs occurred after Full Moon, mean about +3.2 days and median about +6 days. Phase-shift/null controls weaken any claim that the astronomical Full Moon uniquely causes the high.

New-Moon/local-low correspondence is the stronger empirical lead. In 2026, 7 of the first 8 New Moons had a strict local low in T0:T+3 and all 8 were within +/-7 days, while full-history behaviour is regime-dependent.

Nodal strength is not a reliable timing predictor. The exploratory amplitude result is conditional: after a New-Moon-associated low has already formed, stronger New Moons historically correlated with larger subsequent 7/14/21-day rallies. This remains unvalidated prospectively.

Pattern Fit Score remains 100 points: 25 New-Moon low alignment, 25 waxing appreciation, 25 Full-Moon high alignment, 25 waning weakening. Nodal strength and market state are stored separately and do not inflate the score.

Research protocol changes are versioned. Never silently alter lag windows, pivot definitions, scoring weights or strength formula because recent price action did not fit.

9. Frozen September 2026 Protocol

Create an immutable ResearchProtocol record: New Moon 11 Sep 2026 03:27 UTC; test whether a strict local low forms during 11-14 Sep (T0:T+3); New-Moon strength about 79/100 recorded in advance but not evidence that a low must occur; if a qualifying low forms, measure maximum upside exactly 7, 14 and 21 days from the pivot; Full Moon 26 Sep 2026 16:49 UTC; then assess waxing return, cycle-high alignment and website-defined major-high relationship; final review after 30 Sep. Do not modify this protocol in response to outcomes.

10. Data Model

Implement/adapt MySQL tables for: users/admins; conversations; messages; research_briefings; sources; news_items; market_snapshots; lunar_events; natal_events; technical_patterns; experiments; observations; hypotheses; predictions; results/reviews; articles; research_protocols; ai_usage/cost_ledger; scheduled_job_runs; subscriptions/entitlements.

Important timestamps: created_at, observed_at, published_at, result_recorded_at, source_published_at. Historical reconstruction must preserve observation date separately from actual import/publication date.

Add provenance values such as live_site, contemporaneous_chat, research_paper, reconstructed_archive, automated_research. Publicly reconstructed archive posts must not falsely imply that the website itself published them on the historical date.

Once a public prediction is published, its prediction text, criteria and evidence snapshot are immutable. Results/corrections are appended as new records.

11. Private AI Companion

Build an authenticated private conversation/research interface. It should retrieve relevant active experiments, prior predictions/results, frozen research protocols, lunar/natal context, market snapshots and relevant ingested news before responding.

The AI must explicitly distinguish observed facts, sourced news, technical interpretation, lunar/astrological interpretation, hypothesis, prediction and retrospective result.

Provide editorial actions from chat: Create Observation, Create Hypothesis, Create Prediction, Draft Article, Record Result, Attach to Existing Experiment, Archive/Ignore. A simple robust v1 is preferred over a complex workflow.

Use the OpenAI API via the current supported Python SDK. Centralise model selection/configuration. Do not hard-code one expensive model for all tasks.

12. Twice-Daily Researcher

Run private research briefings around 07:00 and 18:00 Europe/London, configurable and triggered by cron/CLI rather than relying on a long-running browser session.

Gather only useful BTC context: current price/24h movement; volume/volatility; major support/resistance and technical structures; ETF/flow data where reliable; liquidations/derivatives where reliable; FOMC/Fed speakers; CPI/PCE/payrolls/GDP; major options expiries; SEC/CFTC/regulation; institutional events; material geopolitical developments; exchange/security events; current Moon phase and FM/NM offset; upcoming frozen lunar windows; notable BTC natal transits; active experiment status.

The briefing is concise and prioritised. End with editorial triage: Nothing material / Watch / Possible story / Experiment update required / Prediction or result requires review.

Scheduled predictable events should be surfaced BEFORE they happen. Unexpected catalysts should be captured after detection with source links and an explanation of why they may matter.

13. News/RSS Layer

News is context, not the site's identity. Ingest configurable RSS feeds and authoritative public sources into MySQL. Store headline, canonical URL, publisher, source publication time, retrieved time, category, short summary, relevance score and dedupe fingerprint.

Suggested categories: Macro/Fed, Regulation, ETF/Institutional, On-chain, Liquidations/Derivatives, Technical, Security/Exchange, Geopolitical, Bitcoin-specific, Other.

Deduplicate syndicated stories. Prefer primary/authoritative sources for factual macro/regulatory claims where possible. Do not republish full copyrighted articles. Public stories should link and attribute sources.

Only 3-5 genuinely important developments should normally appear prominently on the homepage.

14. BTC Natal Chart + Outlooks

Create a prominent Bitcoin Natal Chart section explaining chart assumptions and the experimental nature of the interpretation. Astrology is not established financial forecasting and must not be presented as certainty.

Content types: Daily Outlook, Monthly Outlook, Yearly Outlook, Natal Chart Explained, Transit Calendar.

Daily outlooks are normally generated/published only one day in advance to encourage return visits. They combine natal/transit interpretation, current Moon phase/FM-NM offset, relevant frozen experiment window, technical context, important scheduled events/news and observable levels/conditions that support or weaken the thesis.

Everything is free through 31 Dec 2026. Build entitlement/paywall capability now but leave premium gating disabled by configuration. Historical outlooks remain accessible and can have retrospective result/review sections appended.

15. Public Site

Suggested navigation: Home; Experiment; Observations; Predictions & Results; BTC Natal Chart; Outlooks; Research/Methodology; Important BTC News; About.

Homepage order: experiment hero; BTC Now + Moon Now; Current Experiment; Tomorrow's BTC Natal Outlook; Latest Prediction/Result; Research Journal; 3-5 Important Bitcoin Developments; Methodology/Paper; free account/newsletter CTA.

Use server-rendered semantic pages for SEO where practical. Add metadata, canonical URLs, OpenGraph, article structured data, sitemap.xml, robots.txt, clean slugs and internal links between predictions/results/experiments. Do not generate thin SEO spam.

Use clear labels: OBSERVATION, HYPOTHESIS, PREDICTION, RESULT, RESEARCH UPDATE, EXPERIMENTAL. Visitors must be able to distinguish forward-looking records from retrospective commentary.

16. Experiment Record

Create a reusable public experiment record containing: ID; observation time; Moon event/exact offset; BTC snapshot; natal context; technical context; market/news context; hypothesis; prediction; uncertainty/confidence; objective test

criteria; invalidation criteria; status; result; timing error; subsequent high/low; lessons; sources.

Outcomes may be Consistent, Partially Consistent, Inconsistent, Invalidated or Inconclusive. Do not reduce the project to a simplistic win percentage. Preserve numeric data for later statistical testing.

The original prediction/evidence box is locked after publication. Results appear beneath it with their own timestamp.

17. Seed Archive - August/September 2026

Import the following as DRAFT reconstructed archive records. Verify exact market data before publishing. Never invent missing exact timestamps or prices.

30 Aug: updated exploratory paper; recovered exact website methodology; recent Full-Moon +4.4-day result and limitations.

30 Aug: August 2026 high-fit cycle - pre-waning Pattern Fit about 82.6/100, around 97th historical percentile; 12 Aug New Moon -> 14 Aug local low; roughly 30% rally toward Full-Moon/high area; 28 Aug Full Moon strongly nodally aligned; historical high-fit cycles suggested elevated waning weakness, commonly nearer 10-15% than a crash.

Late Aug/early Sep: prospective FM+4.4 turning-window observation. Record Moon timing as a window/context, not a standalone entry instruction.

Early Sep: possible bearish head-and-shoulders / neckline setup appeared provisionally validated but subsequently failed/reclaimed. Preserve this failure.

Early Sep: sharp BTC rally associated with dovish Fed repricing. Lesson: macro catalysts can overwhelm/reshape lunar and technical setups.

11 Sep: formal New-Moon prospective test begins. Keep the paper's T0:T+3 protocol distinct from the informal private NM+0.4-day timing idea.

11-14 Sep: verify actual low(s) from market data. Private notes observed an initial reaction close to the informal NM+0.4 estimate but a later lower low around NM+4; preserve that distinction rather than cherry-picking.

16 Sep: possible large daily cup-and-handle identified around the mid-\$75k area; approximate major rim/confirmation zone discussed was \$81k-\$83k; classical measured-move discussion around \$100k-\$103k if confirmed.

20-21 Sep: smaller intraday cup-and-handle identified inside the larger structure; approximate micro rim around \$81.3k-\$81.6k and projected move toward \$82k+ resistance.

21 Sep: micro breakout and subsequent major-pattern breakout/confirmation should be recorded using verified market data.

Private trade history, leverage, P&L, stop/TP and exchange screenshots are NOT seed content and remain private.

18. Editorial Integrity

Voice: intelligent, curious, evidence-aware, readable, not sensational. Prefer "consistent with", "corresponded with", "failed", "inconclusive" and "requires further testing" over causal certainty.

Always distinguish astrology/lunar interpretation from established market facts. Technical analysis is probabilistic. News/catalyst claims require sources. Failures and misses are publishable and must not be hidden.

Add a clear general notice that this is an ongoing experiment and not investment advice. The public site must not become a personalised trade-instruction service.

19. AI Cost Control - Mandatory

Because the project remains free through year-end, cost telemetry is mandatory from day one. Store provider, model, task, input/output tokens, estimated cost, timestamp, related briefing/article/conversation and cache status.

Admin dashboard: today spend, month spend, spend by task, spend by model, average Morning Brief cost, Evening Brief cost, outlook cost, article cost and Companion cost.

Configurable budget warnings at 50%, 75% and 90%, plus a hard monthly ceiling. At the ceiling, non-essential AI generation stops while the public site, data collection and admin remain operational.

Pipeline should be deterministic/cheap first: RSS/API ingestion -> dedupe -> rule-based/basic relevance -> database -> higher-quality AI only for scheduled briefs, active experiment analysis, outlooks and editor-approved writing. Cache research and do not repeatedly summarise the same source.

Make model choices configurable in environment/admin settings so cheap models can handle classification/summaries and stronger models can be reserved for synthesis/research.

20. Authentication/Security

Admin and Companion require authentication and server-side authorization. No admin public signup. If subscriber accounts are implemented, keep their auth/roles separate from admin permissions.

Protect AI endpoints with auth/rate limits, validate input, escape output, protect sessions/CSRF as appropriate, never expose API keys client-side, and ensure public serializers cannot leak private conversation/trade fields.

Scheduled jobs must be idempotent and log success/error/retry state.

21. Speed-First Build Order

1) Audit repo and deployment. 2) Git checkpoint. 3) Add regression tests around current lunar/research outputs. 4) Choose minimal Flask/Streamlit structure compatible with LiteSpeed. 5) Add SQLAlchemy/MySQL models and migrations. 6) Implement private/public editorial lifecycle. 7) Build public pages. 8) Build/authenticate private Companion/admin. 9) Encode research protocols/lunar services. 10) Add RSS/news ingestion. 11) Add twice-daily cron jobs. 12) Add natal outlook model/pages. 13) Add AI cost ledger/limits. 14) Import reconstructed seed drafts. 15) Add SEO. 16) Run tests/build/startup checks. 17) Write deployment docs and completion report.

Do not stop after the audit or plan. Implement as much as possible tonight. If a provider credential is missing, build the adapter/configuration and continue with the rest.

22. Tonight's Acceptance Criteria

Aim to finish with: application running locally; legacy lunar methodology still reproducible; MySQL-ready migrations; public homepage/journal/outlook routes; authenticated private admin/Companion; observation/hypothesis/prediction/result CRUD; immutable published predictions; research protocol records; reconstructed archive drafts; RSS ingestion foundation; cron-executable Morning/Evening Brief jobs; BTC natal outlook content model; AI cost ledger and budget controls; free-until-31-Dec entitlement flag; basic SEO; tests for privacy and immutable-publication rules; .env.example; PROJECT_CONTEXT.md; DEPLOYMENT.md.

If all features cannot be completed, leave a precise ROADMAP/TODO ranked by impact. Do not leave the repository in a broken half-migrated state.

23. Completion Report

At the end, report exactly: architecture chosen and why; existing files preserved/refactored; schema/migrations; routes/pages; private Companion status; research/news jobs; lunar/natal implementation; AI model configuration; cost controls; seed content status; environment variables; MySQL setup; LiteSpeed deployment procedure; cron entries; tests and results; known issues; and the shortest next steps to deploy.

The final system should be easy for the owner to continue iterating with Claude in future sessions without re-explaining the product.

Appendix - Non-Negotiable Rules

- Stay Python-first. No unnecessary JavaScript framework migration.
- Production datastore is MySQL; migrations must be versioned.
- Optimise deployment for the existing LiteSpeed environment; document actual assumptions after auditing the repo/host setup.
- Preserve legacy research methodology and add regression tests before major refactors.
- Private chat and trades never publish automatically.
- Published predictions are immutable; results are appended.
- Never fake historical publication dates. Observation date and import/publication date are separate.
- Do not change frozen research rules to improve apparent performance.
- News supports the experiment; it must not dilute the lunar/natal research identity.
- Everything remains free through 31 December 2026, but future entitlement/paywall capability is built now.
- Track AI spend and enforce a configurable hard ceiling.
- Failures, misses and inconclusive results remain part of the public scientific record.
- Build quickly and iterate, but never compromise privacy or research provenance.